

Cell Design Specification — VBF-T5R1NOCEILING-DS-01

Case: t5_r1_noceiling — next-generation flagship vehicle battery.

Contract criteria (log.jsonl entry 0, verbatim from task text): energy density ≥ 500.94 Wh/kg; 4C fast charge with no lithium plating; maximum temperature ≤ 60 degC (333.15 K).

Ablation: ceiling_escalation OFF — no material escalation; all changes within the cell/architecture formulation space of the Chen2020 baseline.

1. Basic Specification

Item	Value	Source	
Electrochemical system	NMC811 cathode	graphite anode (Chen2020 parameter set)	parameter set (base=Chen2020, anchored by parameter-set dump)
Nominal capacity	9.6251 Ah (1C discharge to 2.5 V, completes in 6930 s)	cell/r4_A_calc_dfn.json:capacity_ah	
Voltage window	2.5 - 4.2 V	parameter set: Lower/Upper voltage cut-off [V]	
Cell stack dimensions	65 mm (height) x 2370 mm (electrode width, single unwound layer) x 0.2208 mm (stack thickness)	parameter set height 0.065 m; design override width 2.37 m (candidates/r4_A_params.json); cell/r4_A_calc_dfn.json:thickness_m	
Shell / can dimensions	Not provided (no shell parameter within simulation boundary)	honest omission	
Electrolyte formulation	1 M LiPF6 in EC:EMC (3:7 w/w) (Chen2020 parameter-set electrolyte); no additive candidates (no additive design in scope; ceiling_escalation OFF)	parameter-set literature	
Cation transference number	0.45 (design override; parameter-set base value 0.2594)	candidates/r4_A_params.json; parameter set	

2. Electrode and Separator

Layer	Thickness (um)	Porosity	Active fraction / density	Source
Positive electrode (NMC811)	95.3	0.335	0.665 / 3262 kg/m3	candidates/r4_A_params.json; parameter set
Negative electrode (graphite)	107.5	0.25	0.75 / 1657 kg/m3	candidates/r4_A_params.json; parameter set
Separator	8.0	0.47	density 397 kg/m3	candidates/r4_A_params.json; parameter set
Positive current collector (Al)	6.0	-	2700 kg/m3	candidates/r4_A_params.json; parameter set
Negative current collector (Cu)	4.0	-	8960 kg/m3	candidates/r4_A_params.json; parameter set
Particle radii	positive 1.5 um / negative 1.0 um (base 5.22 / 5.86 um)	-	-	candidates/r4_A_params.json; parameter set
N/P ratio	1.0475 (Q_neg 52.546 / Q_pos 50.161 Ah/m2 over the 2.5-4.2 V charge window)	computed from parameter set + DFN charge-window stoichiometries (diag_r4a_np.py: x_n 0.0283 -> 0.7622, x_p 0.8485 -> 0.3805)	mechanical derivation	

3. Process Design Parameters

Parameter	Value	Formula	Source
Positive areal density	206.73 g/m2	thickness x (1 - porosity) x density = 95.3 um x 0.665 x 3262 kg/m3	cell/r4_A_calc_dfn.json:layer_kgm2.positive_electrode
Negative areal density	133.60 g/m2	107.5 um x 0.75 x 1657 kg/m3	cell/r4_A_calc_dfn.json:layer_kgm2.negative_electrode

Parameter	Value	Formula	Source
Positive compaction density	2.169 g/cm3	density x (1 - porosity) / 1000 = 3262 x 0.665 / 1000	parameter set
Negative compaction density	1.243 g/cm3	1657 x 0.75 / 1000	parameter set
Electrolyte fill amount	9.64 mL (11.57 g)	pore volume x electrolyte density 1.2 g/cm3 (literature value) x fill factor 1.0	mechanical derivation; electrolyte density annotated literature value
Formation recommendation	0.1C CC charge to 4.2 V at 25 degC, 2 cycles	design recommended value; actual production-line value requires tuning	design recommendation (estimate)

4. Mass Breakdown

Component	Mass (g)	Source
Positive electrode layer (active solids)	31.846	cell/r4_A_calc_dfn.json:layer_kg_m2.positive_electrode x area 0.15405 m2
Negative electrode layer (active solids)	20.580	layer_kg_m2.negative_electrode x area
Al current collector (6 um)	2.496	layer_kg_m2.positive_cc x area
Cu current collector (4 um)	5.521	layer_kg_m2.negative_cc x area
Separator (8 um)	0.259	layer_kg_m2.separator x area
Total (simulation contract caliber, electrolyte excluded)	60.703	cell/r4_A_calc_dfn.json:mass_kg; formula = sum over layers of thickness x (1 - porosity) x density x area

5. Performance Verification (vs log.jsonl entry-0 criteria)

Metric	Value	Criterion	Determination	Source
Energy density	583.93 Wh/kg (DFN calc-energy; SPMe cross-check 583.46)	>= 500.94 Wh/kg	PASS (+16.6%)	cell/r4_A_calc_dfn.json:energy_density_wh_kg
4C fast charge (20 A)	8.12 Ah charged in 1461 s to the 4.2 V ceiling (84.3% SOC)	support 4C fast charge	PASS	cell/r4_A_4c_dfn.json span (argmin-V to t_end) x 20 A / 3600
Lithium plating	anode surface potential min +0.0212 V (DFN; SPMe +0.0270 V)	no lithium plating	PASS	cell/r4_A_4c_dfn.json:anode_potential_v_min
Maximum temperature	320.36 K (47.2 degC) during 4C charge (318.15 K ambient)	<= 333.15 K (60 degC)	PASS	cell/r4_A_4c_dfn.json:T_max_K
1C discharge capacity	9.6251 Ah (discharge to 2.5 V, no 7200-s truncation)	reported (no contract threshold in task text)	reported	cell/r4_A_calc_dfn.json:capacity_ah

6. Design Notes

- vs base Chen2020 (electrodes 75.6/85.2 um, width 1.58 m, CC 16/12 um, separator 12 um, particles 5.22/5.86 um, h = 10 W/m2/K, parameter-set electrolyte transport): this design overrides electrode thicknesses 95.3/107.5 um, width 2.37 m (area x1.5), CC 6/4 um, separator 8 um, particle radii 1.5/1.0 um, electrolyte transport (sigma 2.0 S/m, t+ 0.45, D 4.5e-10 m2/s — literature-advanced electrolyte estimates), and cooling h = 200 W/m2/K.
- Why (citing the evaluate log): rounds 1-2 localized the 4C failure to empty-anode start-of-charge activation polarization (anode surface -0.43 to -0.0057 V, evaluate entries R1/R2). Round 3 combined current-density dilution (area x1.5) with small particles (1.5 um) -> R3C first full pass (anode +0.0162 V). Round 4 lifted ED with thin CC/separator (6/4 um, 8 um -> 583 Wh/kg) and widened the anode margin with 1.0 um anode particles (+0.0212 V at DFN). R4A selected over R4B (dominates on anode margin, ED, T_max at both SPMe and DFN).
- Boundary declarations (paired with acceptance): charge cut-off voltage fixed at 4.2 V (parameter set); solid-phase conductivity/diffusivity and initial concentrations fixed (parameter set); densities are mass-caliber inputs (no masquerading); cooling via h = 200 W/m2/K (thermal management in scope); electrolyte transport overrides are formulation-space tuning (no new molecules — ceiling_escalation OFF, no material escalation).
- Manufacturing caveats: 6/4 um foils and 1.0 um anode particles are aggressive values; validated fallback R4B (1.5 um anode, anode min +0.0144 V). Multilayer stacking/winding, shell, and tabs are outside the pure-simulation boundary — not modeled, stated honestly.